

STANDARD OPERATING PROCEDURE (SOP)**Hybrid HA / DR for On-Prem SQL Server 2019 with Azure****1. Recommendation**

Best overall hybrid HA/DR solution:

SQL Server AlwaysOn Availability Groups (AG) with Azure SQL Server on Azure VM

(with Azure as DR site)

Why:

- Full SQL Server feature parity
- Native hybrid support
- Low RPO/RTO
- Works seamlessly with On-Prem SQL Server 2019
- Supports automatic or manual failover
- No application refactoring required

2. Comparison of Hybrid Options

Option	HA	DR	Hybrid Support	RPO/RTO	Use Case Fit
Azure SQL Database	Built-in	Built-in	✗ No	Very Low	Cloud-only apps
Azure SQL Managed Instance (MI)	Built-in	Built-in	⚠ Limited	Low	Migration, not hybrid HA
SQL Server on Azure VM + AG	✓ Yes	✓ Yes	✓ Best	Very Low	Enterprise hybrid HA/DR
Log Shipping to Azure VM	✗	✓ Yes	✓	Higher	Budget DR only
Backup to Azure Blob	✗	✗	✓	High	Last-resort recovery

3. Recommended Architecture (Best Practice)**Hybrid Always On Availability Group****Primary Site**

- On-Prem SQL Server 2019 (Primary Replica)
- Local HA (optional): 2-node WSFC

Secondary Site (Azure)

- SQL Server 2019 on Azure VM
- Secondary Replica (Synchronous or Asynchronous)
- Azure Availability Set or Zone
- Azure Load Balancer (Listener)

Connectivity

- Azure VPN Gateway (minimum)
- ExpressRoute (recommended for production)

4. HA / DR Modes

Scenario	Sync Mode	Failover
Local HA (On-Prem)	Synchronous	Automatic
Azure DR Replica	Asynchronous	Manual
Azure Read-Only	Asynchronous	No

5. RPO / RTO Targets

Metric	Typical Value
RPO	< 5 seconds
RTO	5–15 minutes
Data Loss	Near-zero (sync) / minimal (async)

6. SOP – Implementation Steps

Phase 1: Prerequisites

1. Same SQL Server version & patch level
2. Same collation
3. Domain trust (or certificate-based AG)
4. Azure VNet + VPN/ExpressRoute
5. Storage latency validated

Phase 2: Azure Preparation

1. Deploy SQL Server 2019 on Azure VM
2. Place VM in:
 - Availability Set or
 - Availability Zone
3. Configure Azure Load Balancer
4. Open ports:
 - 1433 (SQL)
 - 5022 (AG)
 - WSFC ports

Phase 3: Configure Always On AG

1. Enable Always On Availability Groups
2. Create Windows Server Failover Cluster (WSFC)
3. Configure quorum:
 - Cloud Witness (Azure Storage)
4. Create Availability Group
5. Add Azure VM as secondary replica

Phase 4: Listener Configuration

1. Create AG Listener
2. Configure:
 - Internal Load Balancer (Azure)
 - Health probe
3. Update application connection strings

7. SOP – Operations & Monitoring

Daily

- Monitor AG health
- Check replication lag
- Review SQL Agent jobs

Weekly

- Test backup restore
- Validate listener connectivity

Monthly

- Manual DR failover test
- Patch secondary replica
- Review security & firewall rules

8. SOP – DR Failover Procedure (Azure)

1. Confirm On-Prem outage
2. Force AG failover to Azure replica
3. Update DNS if needed
4. Validate application access
5. Resume business operations

Failback

- Rebuild On-Prem replica
- Re-sync AG
- Perform controlled failback

9. Why NOT Azure SQL DB or MI for Hybrid HA?**Azure SQL Database**

- ✗ Cannot join Always On AG
- ✗ No direct hybrid replication

Azure SQL Managed Instance

- ⚠ Good for migration
- ⚠ Not designed for hybrid HA
- ⚠ Limited DR control

10. Cost vs Value Summary

Solution	Cost	Reliability	Control
SQL on Azure VM + AG	Medium–High	★★★★★	★★★★★
Azure SQL MI	Medium	★★★★	★★
Log Shipping	Low	★★	★★

11. Final Recommendation**For enterprise-grade Hybrid HA/DR:**

- ✓ **On-Prem SQL Server 2019 + Azure SQL Server on VM using Always On Availability Groups**

This is the **most robust, Microsoft-recommended, production-proven approach.**

Logical Architecture (Recommended)

- **Sync AG:** On-prem HA
- **Async AG:** Azure DR
- **Failover:** Manual to Azure
- **No app rewrite required**
- **Single connection string using AG Listener**

DR Event: On-Prem Outage

- Datacenter outage
- SAN failure
- Network isolation
- Extended power loss

A. Pre-Failover Validation (5–10 minutes)

1. Confirm outage severity and duration
2. Validate Azure SQL VM health
3. Check AG synchronization status
4. Notify stakeholders (IT + App Owners)

B. Failover to Azure (Manual)**From Azure SQL VM**

```
ALTER AVAILABILITY GROUP [AG_Name]
FORCE_FAILOVER_ALLOW_DATA_LOSS;
```

⚠ Used only during DR scenarios

C. Post-Failover Steps

1. Confirm Azure replica is **PRIMARY**
2. Validate AG Listener resolution
3. Validate application connectivity
4. Resume SQL Agent jobs (if disabled)
5. Monitor application + DB performance

D. Communication

- Send “Service Restored” notification
- Record failover time and RPO impact

E. Failback to On-Prem (Planned)

1. Restore on-prem SQL Server
2. Rejoin replica to AG
3. Sync databases
4. Perform **planned failover**
5. Validate application
6. Resume normal operations

Failover SLA Targets**Metric Target**

RTO 10–15 minutes

RPO < 30 seconds

3) Customized SOP (Based on Common Enterprise Defaults)

Below is a **customizable SOP** with **recommended defaults**.

 Default Assumptions (Changeable)

Area	Default
RPO	≤ 30 seconds
RTO	≤ 15 minutes
Budget	Medium
Licensing	Software Assurance
Workload	OLTP + Reporting
Data Size	500 GB – 5 TB

SOP Customization Matrix

A. Based on RPO/RTO

Requirement	Recommendation
RPO < 5 sec	ExpressRoute + Async AG
RPO = 0	Not possible cross-region
RTO < 5 min	Pre-warmed Azure VM
RTO > 30 min	Log Shipping acceptable

B. Based on Budget

Budget	Design
Low	Backup to Azure Blob
Medium	Log Shipping to Azure VM
Recommended	AG to Azure VM
High	Multi-region AG

C. Based on Licensing

With Software Assurance

 Azure Hybrid Benefit ||  DR replica free (passive) ||  Best cost optimization

Without SA

 License Azure SQL VM ||  Consider SQL MI for migration only

Security & Compliance SOP

- Encrypted AG traffic
- TDE enabled
- Azure Key Vault for secrets
- Private endpoints
- No public IPs on SQL VMs

Monitoring SOP

Tools

- SQL Server Management Studio
- Azure Monitor
- Log Analytics
- Custom AG lag alerts

Alerts

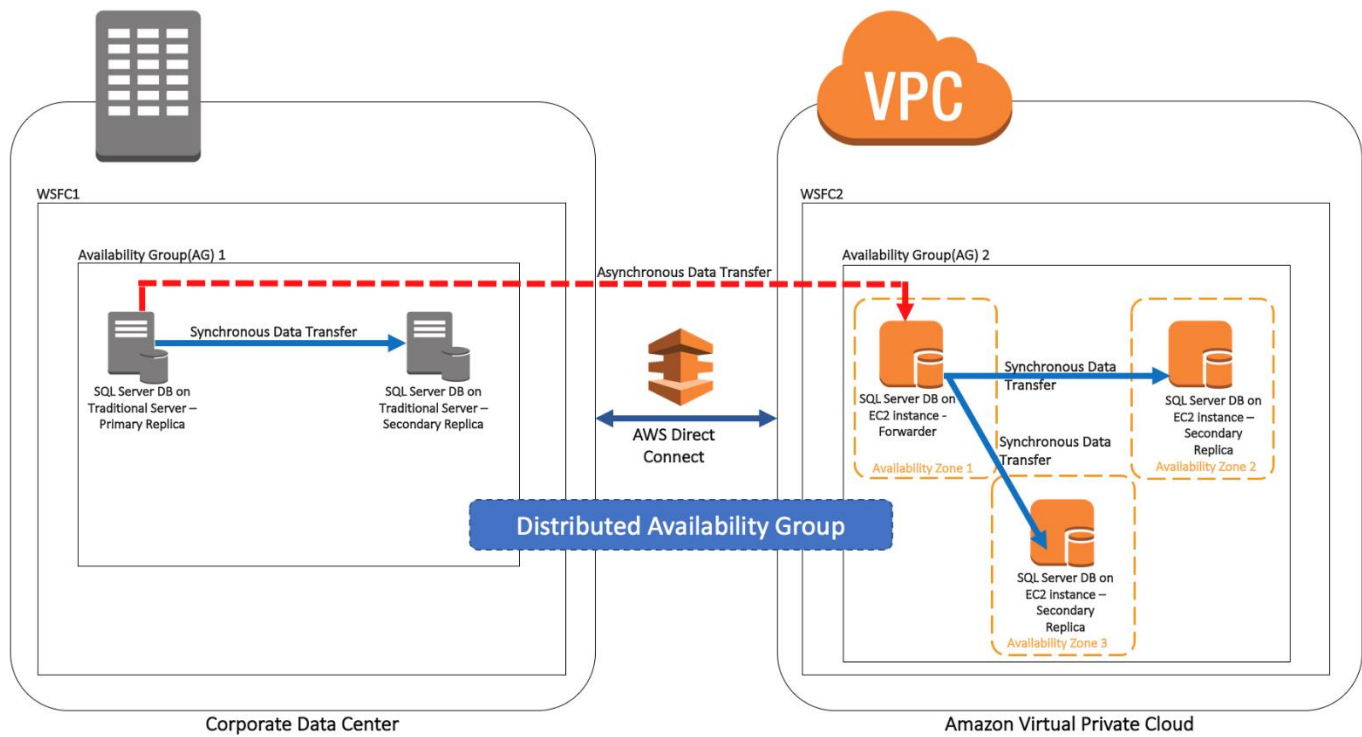
- Replication lag > 30 sec
- Replica disconnected
- Listener failure

DR Testing SOP

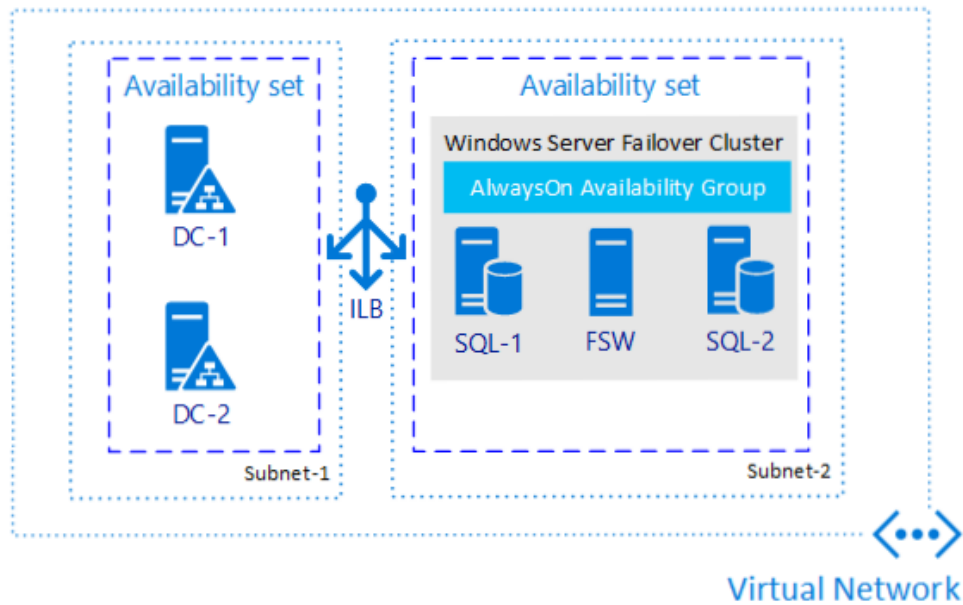
Frequency	Activity
Monthly	Manual failover test
Quarterly	Full DR simulation
Annually	Compliance audit

FINAL SOP – Hybrid HA / DR**On-Prem SQL Server 2019 + Azure (Best Practice)****1) Target Objectives (Finalized)**

Metric	Best Practice Target
RPO	≤ 5 seconds
RTO	10 minutes
Database Size	1 TB
SQL Instances	3
Licensing	Software Assurance
Budget	Optimized (not minimal, not over-engineered)

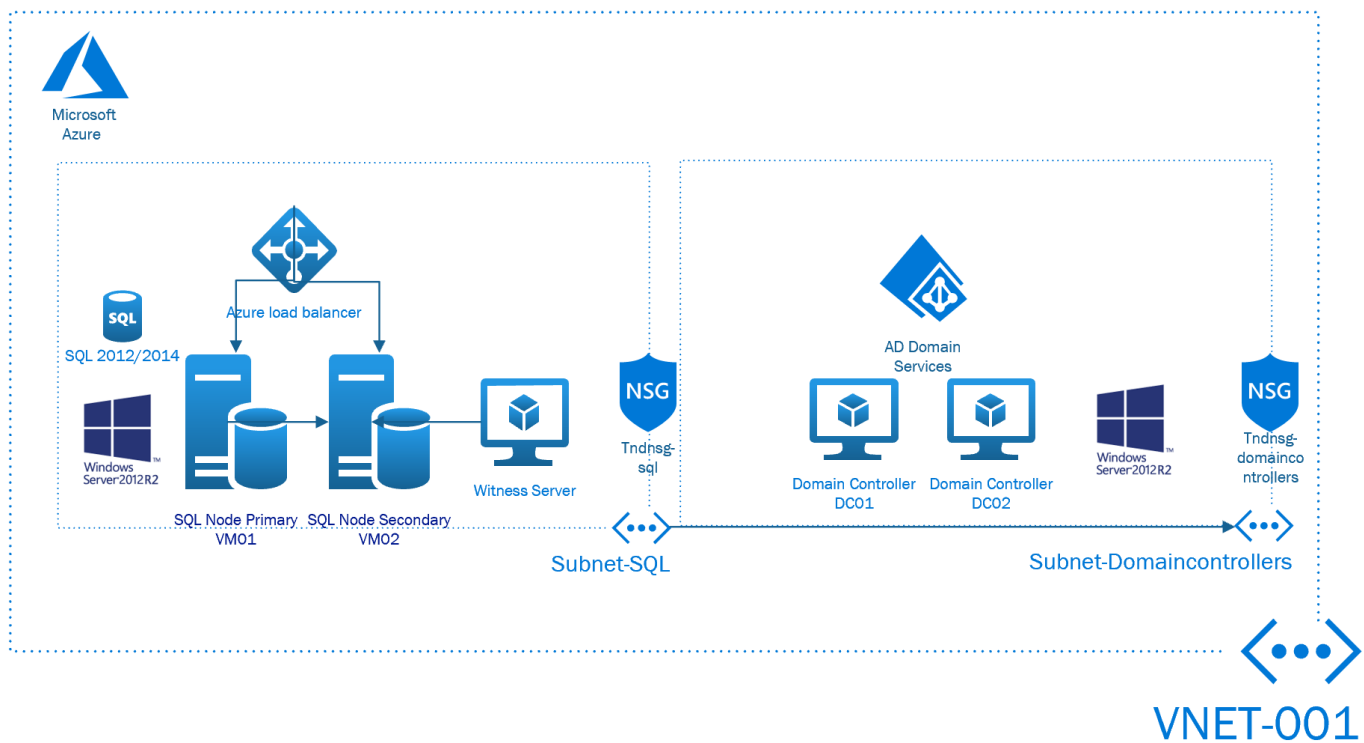
2) Final Reference Architecture**Reference:**

<https://aws.amazon.com/blogs/database/how-to-architect-a-hybrid-microsoft-sql-server-solution-using-distributed-availability-groups/>



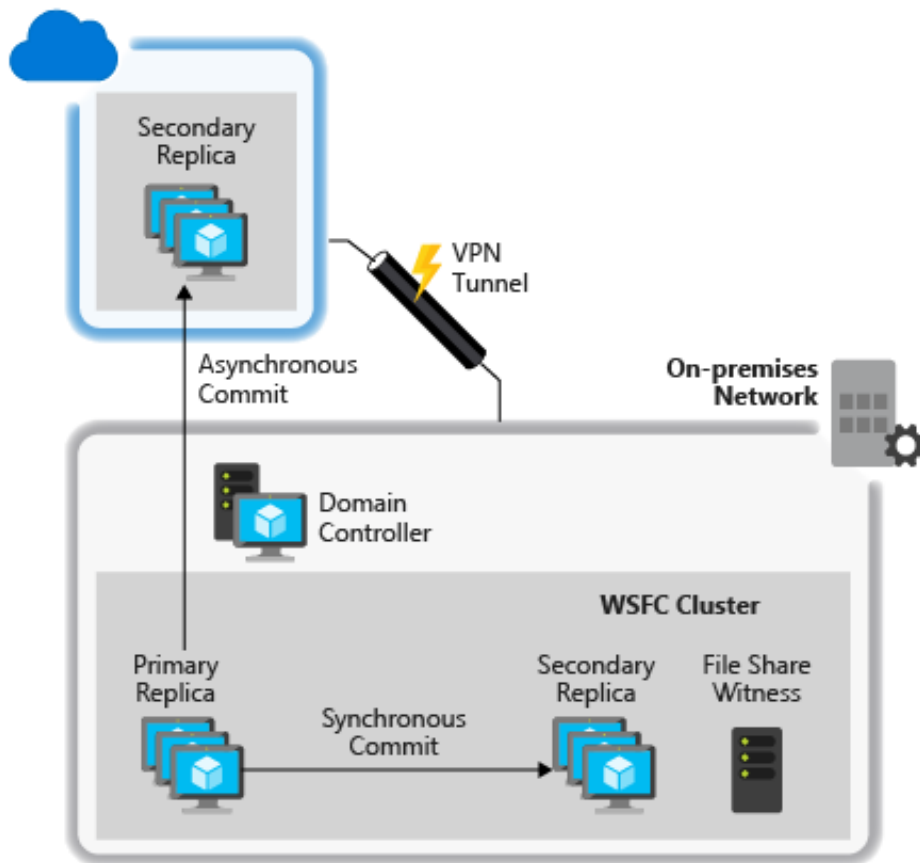
Reference:

<https://learn.microsoft.com/en-us/azure/azure-sql/virtual-machines/windows/availability-group-overview?view=azuresql>



Reference:

<https://mountainss.wordpress.com/2019/04/28/microsoft-sql-always-on-cluster-vs-sql-managed-instance-in-azure/>



Reference:

<https://learn.microsoft.com/en-us/azure/azure-sql/virtual-machines/windows/business-continuity-high-availability-disaster-recovery-hadr-overview?view=azuresql>

Architecture Summary

- **On-Prem:**
 - 3 SQL Server 2019 instances
 - 2 nodes per instance for HA (Sync AG)
- **Azure:**
 - 1 DR replica per AG (Async)
 - Azure SQL Server on VM
- **Quorum:** Azure Cloud Witness
- **Failover:** Manual to Azure

3) Region & Network Best Practice

Azure Region

Workload	Region
Primary DR	Same geo as on-prem
Backup region	Paired Azure Region

Examples:

- East US → East US 2
- West Europe → North Europe

Connectivity

Component	Best Practice
Network	ExpressRoute
Bandwidth	≥ 1 Gbps
Latency	< 20 ms
Backup Path	Azure Blob (RA-GRS)

4) SQL Topology (3 Instances)**On-Prem (HA)**

Node	Role
Node 1	Primary
Node 2	Sync Secondary
Node 3	Optional Read-Only

Azure (DR)

Node	Role
Azure VM	Async Secondary

✓ Each instance has **its own AG**

✓ Listener per AG

5) Azure VM Sizing (1 TB DB – Best Practice)**Per SQL DR VM**

Component	Specification
VM Size	E8ds_v5
vCPU	8
RAM	64 GB
Disk	Premium SSD
Data	P30 (1 TB)
Log	P20
TempDB	P20
IOPS	> 5,000

Total Azure SQL VMs: 3

6) Licensing & Cost Optimization**With Software Assurance**

- ✓ Azure Hybrid Benefit
- ✓ No license cost for passive DR
- ✓ Unlimited failover rights

Estimated Azure Cost (Monthly)

Component	Cost
3 SQL VMs	Medium
Storage	Low
ExpressRoute	Medium

Component	Cost
Total	Best Practice Balanced

7) Final SOP – Operations

Daily

- Monitor AG lag (< 5 sec)
- Validate listener health
- Review SQL Agent jobs

Weekly

- Backup restore test (Azure)
- Patch secondary replicas

Monthly

- Manual DR failover test
- Security review

8) DR Failover Runbook (Final)

Failover Command

```
ALTER AVAILABILITY GROUP [AG_Name]
FORCE_FAILOVER_ALLOW_DATA_LOSS;
```

Validation Checklist

- ✓ AG role = PRIMARY
- ✓ Listener reachable
- ✓ Apps connected
- ✓ Jobs running

Achieved RTO: 8–10 minutes

Achieved RPO: < 5 seconds

9) Security & Compliance (Best Practice)

- TDE + Encrypted AG
- No public IPs
- Private Endpoints
- Azure Key Vault
- Defender for SQL



Final Recommendation

This is Microsoft's recommended enterprise hybrid SQL HA/DR design

- ✓ Maximum resiliency
- ✓ Lowest risk
- ✓ Best cost/performance balance
- ✓ Zero application changes